令和7年度名古屋大学大学院工学研究科

外国人留学生入学者選抜試験

博士課程前期課程

ACADEMIC YEAR 2025
EXAMINATION FOR FOREIGN STUDENTS
FOR ADMISSION TO GRADUATE STUDY PROGRAMS

Dept. Materials Design Innovation Eng.,
Dept. Materials Process Eng.,
Dept. Chemical Systems Eng.
The Graduate School of Engineering
Nagoya University

【語学：英語】

【Language Test: English】

１０:３０～１２:００

- 問題1冊子、答案用紙は全てを提出すること.
- Give hand all problem sheets and answer sheets.

【1】 次の文章を読み，各設問について解答せよ．Read the following passage and answer to each question.

この部分につきましては、著作権の都合により公開いたしません.

1) 次の質問に答えよ．Answer the following questions.

(1) Write down two examples of advanced materials mentioned in the passage?

(2) Why are solid-state electrolytes important for batteries and fuel cells?

(3) What is one benefit of using carbon fiber composites in energy systems?

(4) What challenge is associated with perovskite solar cells?

(5) According to the passage, what additional effort is needed to ensure a sustainable energy transition?

2) 最も適切な単語を選択してください．Choose the most appropriate word to complete each sentence based on the passage.

(1) Perovskite solar cells are popular because of their high (a) cost / (b) efficiency / (c) weight.

(2) Carbon fiber composites help improve energy performance by (a) increasing / (b) reducing / (c) maintaining structural weight.

(3) Recycling rare metals is crucial to ensure (a) efficiency / (b) sustainability / (c) durability in the use of advanced materials.

3) 空欄に動詞または単語の正しい形を入力してください．Fill in the blanks with the correct form of the verb or word.

(1) If the stability of perovskite materials improves, they _____ more commonly used in solar energy systems.

(2) Solid-state electrolytes _____ the safety and efficiency of batteries.

(3) Advanced materials, such as carbon fiber, _____ an essential role in building energy-efficient infrastructure.

【2】 次の設問の空欄について，4つの選択肢の中から最も適切な語句を選びなさい．そして，解答用紙に(a)～(d)の記号を記入しなさい．Choose the best word or phrase from among the four choices for each blank. Then write your answers with the symbols (a) through (d) in the answer sheet.

(1) When preparing a manuscript, it is important to ensure that all references are properly _____ according to the journal's guidelines.
(a) summarized (b) cited (c) highlighted (d) analyzed

(2) The authors were asked to provide a detailed _____ of the experimental methods used in their study.
(a) procedure (b) introduction (c) conclusion (d) acknowledgment

(3) In order to submit the paper, the corresponding author must create an account on the journal's _____ system.
(a) submission (b) editing (c) review (d) plagiarism

(4) The _____ review process ensures that the manuscript meets the quality standards of the journal.
(a) peer (b) self (c) blind (d) editorial

(5) Tables and figures should be placed in the manuscript where they are first _____ in the text.
(a) mentioned (b) repeated (c) rejected (d) ignored

(6) The authors received a request to revise their paper and address the reviewers' _____.
(a) suggestions (b) opinions (c) instructions (d) restrictions

(7) During the presentation, the speaker emphasized the need for careful _____ of the data before drawing conclusions.
(a) exaggeration (b) validation (c) fabrication (d) suppression

(8) The paper was rejected due to a lack of _____ novelty in the research findings.
(a) sufficient (b) immediate (c) irrelevant (d) optional

(9) Presenters at the conference were advised to stay within the allocated _____ time.
(a) meeting (b) presentation (c) preparation (d) recording

(10) The authors were asked to include a conflict of interest _____ in their manuscript.
(a) statement (b) notice (c) approval (d) resolution

【3】 以下の(1)から(3)に示すテーマに関するエッセイを書きなさい．Write essays on the following topics (1) through (3).

(1) 研究者になって何を成し遂げたいか 300 文字以内で説明をしてください．Describe what do you want to accomplish as a researcher? in 300 words or less.

(2) (1)で回答したことを実現するために何をすればよいと考えるか 150 文字以内で説明をしてください．Describe in 150 characters or less what you think should be done to achieve what you answered in (1).

(3) あなたがこれから行う研究分野の世界的な先行研究の状況について 150 文字以内で説明をしてください．Describe the current status (previous research all over the world) of the research field you will do in 150 words or less.

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The Graduate School of Engineering
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【基礎：数学，物理学，化学】

【Mathematics, Physics, Chemistry】

１３：００～１６：００

- 問題1冊子、答案用紙（下書用紙を含む)は全てを提出すること.
- Give hand all problem sheets and answer sheets including rough paper at the end of examination

【数学】　【Mathematics】

【１】 Differentiate the following function y with respect to x.

次の関数 y を x について微分せよ.

$$y=(1+x)^x \quad (x>-1)$$

【２】 Let n be a natural number, and $I_n=\int(x^2+2x+3)^{-n}dx$ $(n\geq 1)$.

n を自然数, $I_n=\int(x^2+2x+3)^{-n}dx$ $(n\geq 1)$ とする.

(1) Derive the following recurrence formula.

次の漸化式を導出せよ.

$$4nI_{n+1}-(2n-1)I_n=\frac{x+1}{(x^2+2x+3)^n} \quad (n\geq 1)$$

(2) Find I_2.

I_2 を求めよ.

【３】 Find a general solution of each of the following ordinary differential equations.

次の常微分方程式の一般解を求めよ.

(1) $\quad x\dfrac{dy}{dx}=y+x^2\cos x$

(2) $\quad \dfrac{d^2y}{dx^2}+3\dfrac{dy}{dx}+2y=12x^2$

【４】 Find a unit normal vector of the surface $2x^2+4y^2+3z^2=30$ at the point P $(-1,2,-2)$.

曲面 $2x^2+4y^2+3z^2=30$ の点 P $(-1,2,-2)$ における単位法線ベクトルを一つ求めよ.

【５】 Let $f(x,y,z)=\log(x^2+y^2+z^2)$, where x, y, z are Cartesian coordinates. Find the divergence of the gradient of f.

$f(x,y,z)=\log(x^2+y^2+z^2)$ とする. ただし, x,y,z はデカルト座標である. f の勾配の発散を求めよ.

【物理学】　　【Physics】

【１】As illustrated in Figure 1, a mass point with mass m is suspended from a spring with a spring constant k and a natural length L. The mass point is pulled down from its natural length by A and subsequently removed gently, at which point the mass point begins to oscillate vertically. The gravitational acceleration is designated as g. The x-axis is defined as the vertical direction, with positive values indicating downward movement. The origin of the x-axis is set at the point where the spring is suspended from the ceiling, as shown in Figure 1.

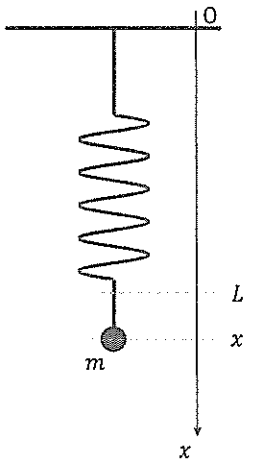


Figure 1

図１のように質量 m の質点がバネ定数 k，自然長 L のバネにつるされている．質点を自然長から A だけ下に引っ張り，静かに手を離したところ，質点は上下に振動をはじめた．重力加速度は g とする．鉛直方向に x 軸をとり，下向きを x 軸の正方向とする．x 軸の原点 O は図１のようにバネがつるされている天井の位置とする．

(1) Express the equation of motion for the mass point.

質点の運動方程式を表せ．

(2) Solve the equation of motion in (1) to express the position x of the particle as a function of time t. Describe the process of solving the equation of motion in (1).

(1)の運動方程式を解くことにより，質点の位置 x を時刻 t の関数として表せ．(1)の運動方程式を解く過程も記述すること．

（次のページに続く　Continue to the next page）

(3) Show that the average value of kinetic energy and the average value of potential energy are equal during one period of oscillation of the mass point.

質点の振動の１周期において，運動エネルギーの平均値と位置エネルギーの平均値が等しいことを示せ．

(4) As shown in Figure 2, an object with mass M is suspended by three springs A, B, and C with spring constants k_1, natural lengths L_0. When the object is at rest, the lengths of springs A, B, and C are all L_1. Find the period of the vertical motion of the object using L_0, L_1 and g, after it is slightly displaced in the vertical direction and then released.

図２のようにバネ定数 k_1，自然長 L_0 の３つのバネ A，B，C により質量 M の物体がつるされている．この物体が静止しているとき，ばね A，B，C の長さはすべて L_1 であった．物体をわずかに鉛直方向にずらして静かに手を離したあとの物体の上下運動の周期を L_0，L_1 および重力加速度 g を用いて表せ．

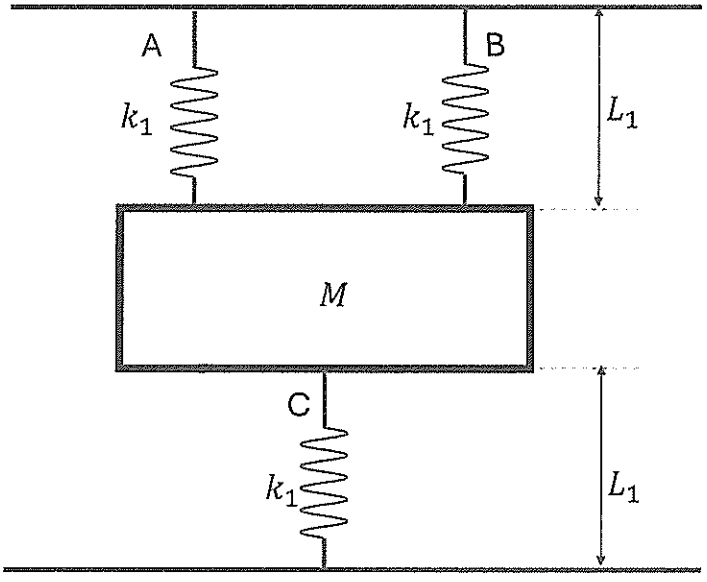


Figure 2

【２】We consider the conditions for a uniform sphere with mass M and radius r, which is rolling at a speed v_0, to climb up a step with a height h ($h < r$), as shown in Figure 3. The edge A of the step is completely rough, and the sphere rises up the step without losing contact with A after colliding with A in a completely inelastic collision. At this time, the impulsive force that the sphere receives from A is assumed to have no moment around A. The moment of inertia of the sphere about the axis passing through the center of the sphere is $\frac{2}{5}Mr^2$. The acceleration of gravity is g.

図 3 のように速度 v_0 で転がってきた質量 M，半径 r の一様な球が高さ h ($h < r$) の段差の上にあがるための条件について考察する．段差のふち A は完全に粗で，球は A と完全非弾性衝突した後，A との接触を失わずに段差に上がる．このとき，球が A から受ける撃力は A のまわりのモーメントをもたないものとする．また，球の中心を通る軸に対する球の慣性モーメントは $\frac{2}{5}Mr^2$ とする．重力加速度は g とする．

(1) Show the angular momentum that the sphere has before it reaches the step.

球が段差に到達する前に持っている角運動量を示せ．

(2) Show the kinetic energy that the sphere has before it reaches the step.

球が段差に到達する前に持っている運動エネルギーを示せ．

(3) Express the angular velocity ω_0 around A of the sphere immediately after it collides with A using r, h and v_0. The angular momentum of the sphere is assumed to be conserved immediately before and after the sphere collides with A.

（次のページに続く　Continue to the next page）

球が段差に衝突した直後の球のAの周りの角速度ω_0をr，h，v_0を用いて表せ．球がAに衝突する直前と直後で球の角運動量は保存するものとする．

(4) Let θ be the angle between the radius OA of the sphere and the horizontal plane. The law of conservation of energy holds immediately after the collision. When the sphere is climbing up the step without leaving A, express the square of angular velocity ω^2 of the sphere at θ using ω_0, r, θ, h, and g. The condition for the sphere to climb up the step is $\omega^2 > 0$ at $\theta = 90°$.

球の半径OAと水平のなす角をθとする．衝突直後から，球がAから離れることなく段差を上っているときにエネルギー保存則が成り立つとして，θにおける球の角速度の2乗ω^2をω_0，r，θ，h，gを用いて表せ．球が段差を上がるための条件は$\theta = 90°$において$\omega^2 > 0$である．

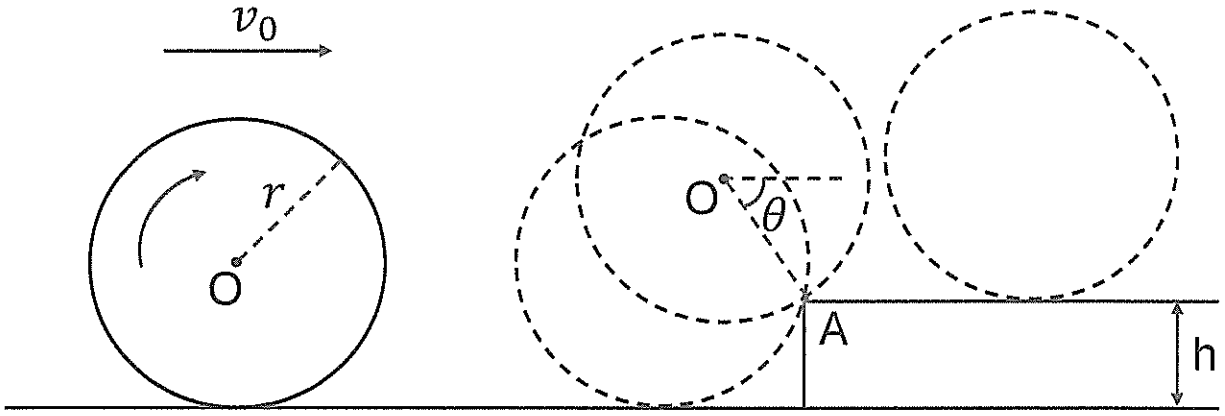


Figure 3

【化学】　　【Chemistry】

【１】For each of the following molecules (1)-(8), answer the number of σ (sigma) bonds, the number of π (pi) bonds, and the type of the hybrid orbitals of the underlined atoms, respectively.
以下の各分子(1)～(8)について，σ（シグマ）結合の本数，π（パイ）結合の本数，下線の原子の混成軌道の種類をそれぞれ答えよ．

(1) $\underline{C}H_4$,　(2) $\underline{N}H_3$,　(3) $\underline{C}O_2$,　(4) $\underline{C}O$,
(5) $H_2\underline{O}$,　(6) $\underline{B}F_3$,　(7) $\underline{P}F_5$,　(8) $\underline{S}F_6$

【２】Identify the Lewis acid and Lewis base in the reaction, $NH_3 + BF_3 = NH_3BF_3$, and explain the reason for the decision in a few sentences or less.
反応 $NH_3 + BF_3 = NH_3BF_3$ におけるルイス酸とルイス塩基を答えて，その判断理由を数文以内で説明せよ．

【３】Determine the equilibrium constant (K) for the reaction, $A^{3+}(aq) + B(s) = A(s) + B^{3+}(aq)$, where A and B are metal elements, under the following conditions and provide the answer in natural logarithm ($\ln K$).
金属元素 A および B の反応 $A^{3+}(aq) + B(s) = A(s) + B^{3+}(aq)$ における平衡定数（K）を以下の条件で求めて，自然対数値（$\ln K$）で答えよ．

標準電極電位（対標準水素電極）
Standard electrode potential (vs. standard hydrogen electrode)

$A^{3+}(aq) + 3e^- = A(s)$　$+0.58$ V
$B^{3+}(aq) + 3e^- = B(s)$　-0.25 V

温度 Temperature　3.0×10^2 K
気体定数 Gas constant　8.3 J K^{-1} mol^{-1}
ファラデー定数 Faraday constant　9.6×10^4 C mol^{-1}

（問題文は次ページに続く．The questions continue on the next page.）

【4】In the irreversible first-order reaction of the decomposition of a compound C, 10% decomposition occurs in 500 s at 400 K, while 80% decomposition occurs in 20 s at 500 K.
化合物 C の不可逆な 1 次分解反応において，400 K では 500 秒間で 10%が分解し，500 K では 20 秒間で 80%が分解する．

$\ln x = 2.3 \log_{10} x$, $\ln 2 = 0.7$, $\ln 3 = 1.1$, $\ln 5 = 1.6$, $\ln 7 = 1.9$

気体定数 Gas constant　8.3 J K^{-1} mol^{-1}

(1) Determine the reaction rate constants (k) at 400 K and 500 K ($k(T = 400\ \mathrm{K})$ and $k(T = 500\ \mathrm{K})$), respectively.
400 K および 500 K における反応速度定数（$k(T = 400\ \mathrm{K})$ と $k(T = 500\ \mathrm{K})$）をそれぞれ求めよ．

(2) Determine the activation energy (E_a), assuming that the dependence of E_a on temperature is negligible.
活性化エネルギー（E_a）を求めよ．ただし，活性化エネルギーの温度依存性は無視できるものとする．

(3) Determine the ratio of the half-life ($t_{1/2}$) at 250 K to that at 500 K (ratio, $r = \frac{t_{1/2}(T=250\ \mathrm{K})}{t_{1/2}(T=500\ \mathrm{K})}$) and provide the answer in natural logarithm ($\ln r$).
半減期（$t_{1/2}$）が 250 K では 500 K のときに比べて何倍になるか，半減期の比（$r = \frac{t_{1/2}(T=250\ \mathrm{K})}{t_{1/2}(T=500\ \mathrm{K})}$）を求めて，自然対数値（$\ln r$）で答えよ．

英語：出題意図
読解力、文法、英作文等に関する知識を問う。

数学：出題意図
微分・積分学、常微分方程式、線形代数、ベクトル解析に関する知識を問う。

物理学：出題意図
質点および剛体の力学に関する知識を問う。

化学：出題意図
化学結合、化学平衡、反応速度等に関する知識を問う。

令和6年度実施　博士前期課程外国人留学生入学試験口頭試問

専攻名	材料デザイン工学，物質プロセス工学，化学システム工学

１人あたりの時間	10分程度
方法	対面
内容	専門分野に関する試問
事前課題	（なし）

備考

（なし）